DEVELOPMENT DOCUMENTATION

4. Operations Runbook

Server, deployment, users and troubleshooting

Dynamic Auctioneers Marketing Platform
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Everything needed to run, deploy and troubleshoot the live platform.

4.1 The server

Host	Hostinger VPS, Ubuntu 24.04 LTS
IP	`46.202.175.127`
Live URL	`https://46.202.175.127.nip.io` — **temporary**, see 4.7
SSH	`ssh root@46.202.175.127`
Application directory	`/opt/da-marketing` (a clone of the repository)
Runs as	systemd service `da-marketing` — one uvicorn process on `127.0.0.1:8000`, behind nginx, as the non-root user `dauction`
Verified	Responding HTTP 200 on 20 August 2026

Layout on the box

THING	PATH
Code	`/opt/da-marketing`
Virtual environment	`/opt/da-marketing/venv`
Secrets	`/opt/da-marketing/.env` — mode `600`, owner `dauction`
Database	`/opt/da-marketing/engine.db` — mode `600`
Property data, uploads, artifacts	`/opt/da-marketing/DP<dp>/...`
systemd unit	`/etc/systemd/system/da-marketing.service` (copy in `docs/deploy/`)
nginx site	`/etc/nginx/sites-available/da-marketing` (copy in `docs/deploy/`)
TLS certificate	`/etc/letsencrypt/live/46.202.175.127.nip.io/` — auto-renews

4.2 Everyday commands

```
systemctl status da-marketing        # is it running
systemctl restart da-marketing       # restart after a config or code change
journalctl -u da-marketing -f        # live logs
journalctl -u da-marketing --since "10 min ago"
```

4.3 Deploying an update

Push to GitHub, then from the repository root on the development machine:

```
./scripts/deploy.sh
```

It SSHes in, pulls `main` fast-forward only, reinstalls dependencies **only if `requirements.txt` changed**, restarts the service, and confirms the site is back at HTTP 200. The SSH key must be loaded in the agent (`ssh-add --apple-use-keychain`).

By hand, the same thing:

```
ssh root@46.202.175.127
cd /opt/da-marketing
sudo -u dauction git pull
sudo -u dauction ./venv/bin/pip install -q -r requirements.txt   # only if deps changed
systemctl restart da-marketing
```

`engine.db` and the `DP<dp>/` folders are untracked, so a pull never touches data.

4.4 Users and roles

ROLE	CAN DO
`admin`	Everything, including the Settings screens that hold API credentials
`marketing`	Run properties end to end — intake, photographs, edits, regeneration
`approver`	Sign the gates. Can act from the approval email without logging in

The split exists so operational staff run everything but **never touch credentials** (D34).

Adding a user: log in as `admin@dynamicauctioneers.co.za` and use the Settings screen. The admin temporary password is printed **once** to the journal on first boot:

```
journalctl -u da-marketing | grep -i "temp password"
```

Change it on first login. A forced first-login password change is a recommended improvement that has not been built.

Sign-in hardening in place (D44): bcrypt with per-password salts; HMAC-signed HttpOnly SameSite=lax session cookies; a brute-force throttle keyed on the account email (8 failures in 5 minutes → 15 minute lockout, tunable with `LOGIN_THROTTLE_MAX_FAILS`, `LOGIN_THROTTLE_WINDOW`, `LOGIN_THROTTLE_LOCKOUT`); constant-time login so response timing does not reveal which emails are registered; a generic error message; per-request user reload so a deleted or role-changed account takes effect immediately; and failed logins logged with email and source IP, never the password.

The throttle is keyed on the **account**, not the client IP, because behind the nginx proxy the peer is localhost and `X-Forwarded-For` is attacker-controlled. The accepted trade-off is that someone who knows an email can nuisance-lock it; the window self-clears and the signed email approval links keep working.

4.5 Developer SSH access

Full instructions, including key generation on macOS and Windows, are in `docs/SERVER-ACCESS.md`. In short: each developer generates their **own** key pair and sends only the **public** half (`~/.ssh/id_ed25519.pub`, one line).

```
ssh root@46.202.175.127
echo "ssh-ed25519 AAAA... theirname-dev" >> ~/.ssh/authorized_keys
```

Revoke by deleting their line. One key per person, no shared secrets. Private keys are never emailed, pasted into chat, or committed.

4.6 Environment variables

Held in `/opt/da-marketing/.env` (mode 600). The template is `.env.example` in the repository, which is kept blank.

Core

VARIABLE	PURPOSE
`ANTHROPIC_API_KEY`	Extraction, verification and copy. Without it the system falls back to templates and deterministic checks rather than failing
`ENGINE_DB`	Database path. Defaults to `./engine.db`
`APP_SECRET`	Signing secret for sessions and approve-by-email tokens. Generated at provisioning
`ENGINE_ALLOW_INSECURE_COOKIE`	**Local development only.** Set to `1` to stop the session cookie being marked Secure. Leave unset in production

> **The session cookie is Secure by default.** Setting `ENGINE_ALLOW_INSECURE_COOKIE` logs a loud warning on every boot, so a misconfigured production box is obvious in the journal.
>
> **Documentation drift, worth knowing:** `.env.example`, `docs/deploy/DEPLOY.md` and decision D44 all refer to `ENGINE_HTTPS=true` as the thing that marks the cookie Secure. **No code reads `ENGINE_HTTPS`.** The behaviour is correct and secure by default — the variable is simply vestigial and the older documents are stale. Do not rely on setting it.

Rendering

VARIABLE	PURPOSE
`ENGINE_RENDERER`	`html` (default), `canva`, or `mixed`
`ENGINE_PDF_EXPORT`	PDF export control
`ENGINE_AI_CACHE`	Location of the content-addressed cache for the paid calls
`EXTRACT_PDF_MODE`	`pdf` (default, native document blocks) or the text fallback
`EXTRACT_PACE_SECONDS`	Call pacing. A legacy workaround for the old 10k tokens/minute tier; not needed now

Distribution — GoHighLevel

VARIABLE	PURPOSE
`GHL_API_TOKEN`	The `pit-...` token. Normally entered in the platform Settings screen and stored in the database, not here
`GHL_LOCATION_ID`	Location
`GHL_USER_ID`	User
`GHL_ACCOUNT_MAP`	JSON mapping channel to account id, e.g. `{"facebook":"acc_id","instagram":"acc_id"}`
`GHL_POST_STATUS`	**The guard rail.** `draft` overrides any per-post choice made in the UI

Canva (optional)

`CANVA_CLIENT_ID`, `CANVA_CLIENT_SECRET`, `CANVA_REFRESH_TOKEN`, `CANVA_REDIRECT_URI`, `CANVA_TEMPLATE_MAP`, `CANVA_STATE_FILE`.

Only needed with `ENGINE_RENDERER=canva` or `mixed`. `CANVA_TEMPLATE_MAP` is JSON, either flat (one design set) or named sets, where the **first set is the default** and defines which formats route through Canva. Refresh-token rotation is persisted to the state file. Authorisation helpers are in `scripts/canva_authorize.py` and `scripts/canva_reauth.py`.

Microsoft Graph (optional, not on the critical path)

`MS_GRAPH_CLIENT_ID`, `MS_GRAPH_CLIENT_SECRET`, `MS_GRAPH_TENANT_ID`, `MS_GRAPH_SITE_ID`, `MS_GRAPH_DRIVE_ID`.

4.7 Cutting over to the real domain

The platform should be on `marketing.dynamicauctioneers.co.za`. The nip.io hostname is a stand-in. This needs whoever holds the DNS.

1. Add an **A record** for the subdomain pointing at `46.202.175.127`. Wait for it to resolve: `dig +short marketing.dynamicauctioneers.co.za`.

2. On the server:

```
sed -i 's/server_name .*/server_name marketing.dynamicauctioneers.co.za;/' \
  /etc/nginx/sites-available/da-marketing
nginx -t && systemctl reload nginx
certbot --nginx -d marketing.dynamicauctioneers.co.za --redirect \
  --non-interactive --agree-tos -m keegs.haumann@gmail.com
```

3. Nothing else changes. The old nip.io certificate can be left to expire.

4.8 Rebuilding the box from scratch

Install `python3-venv python3-pip nginx certbot python3-certbot-nginx git`; clone the repository to `/opt/da-marketing`; create the virtual environment and `pip install -r requirements.txt`; run `playwright install chromium` and `playwright install-deps chromium`; copy `.env` across and append `ENGINE_DB`, a generated `APP_SECRET`; create the `dauction` system user and `chown -R dauction:dauction /opt/da-marketing`; install the systemd unit and nginx site from `docs/deploy/`; set the `output_root` database setting to `/opt/da-marketing`; then run `certbot --nginx`.

4.9 Troubleshooting

SYMPTOM	WHERE TO LOOK
Site down	`systemctl status da-marketing`, then `journalctl -u da-marketing --since "10 min ago"`
A property stuck in a state	The job worker. Failed jobs park the record with the raw model output attached — check the journal for the job kind (`extract`, `verify`, `render`, `post`)
Extraction returning nothing	Check `ANTHROPIC_API_KEY` is present. Without it the system degrades to deterministic checks and template copy rather than erroring
Adverts render but the PNG does not	Playwright Chromium is missing. `playwright install chromium` in the virtual environment
Posts not appearing on a page	Check `GHL_POST_STATUS` — if it is `draft`, everything lands as a draft in the planner regardless of the UI choice. This is intended
Cookie warnings on boot	`ENGINE_ALLOW_INSECURE_COOKIE` is set. It should not be, in production
OTP or levy figures read as missing	Should no longer happen (D77). If it does, confirm PyMuPDF is installed — the readers no longer use `pdftotext`, which the server does not have

4.10 Backups — the open gap

Not yet configured. This is the most significant operational gap in the system.

The database is small and holds every record, approval and audit event. The `DP<dp>/` folders hold the source documents, photographs and artifacts.

Two options, either acceptable:

- Hostinger's paid daily VPS auto-backup — simplest.
- A cron job running `sqlite3 engine.db ".backup ..."` and pushing the copy off-box, for example rclone to Backblaze B2 or to OneDrive.

Until one is in place, a lost VPS is a lost system.